

3. Which of the following memory allocation scheme suffers from External Fragmentation?

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Pick **ONE** option

☐ Paging

☐ Swapping

☒ Segmentation

☐ Pure Demand Paging

Clear Selection

14. The potential challenge of using a monolithic architecture

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Pick **ONE** option

- ☐ Complex deployment process
- ☐ Improved scalability
- ☒ Tight coupling between components
- ☐ High maintenance overhead"

Clear Selection

21. What is the broadcast address for the subnet 172.16.10.128/27?

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Pick **ONE** option

☒ 172.16.10.159

☐ 172.16.10.191

☐ 172.16.10.224

☐ 172.16.10.255

Clear Selection

19. You are designing a REST API for a blogging platform. What URI should you use for creating a new blog post?

You are designing a REST API for a blogging platform. What URI should you use for creating a new blog post?

Pick **ONE** option

☐ /blog/posts/new

☐ /blog/create/post

☒ /blog/posts ✓

☐ /blog/posts/create

Clear Selection

4. Which software loads an Operating System in memory?

Which software loads an Operating System in memory?

Pick **ONE** option

☐ Compiler

☒ Bootloader ✓

☐ Kernel

☐ Device Driver

Clear Selection

7. In an AVL tree, what is the purpose of performing rotations?

In an AVL tree, what is the purpose of performing rotations?

Pick **ONE** option

☐ To keep the tree complete

☐ To ensure that all leaf nodes have the same depth

☒ To maintain balance of the tree to ensure $O(\log n)$ height ✓

☐ To facilitate easy in-order traversal

Clear Selection

11. Which of the following commands are a part of Data Control Language?

Which of the following commands are a part of Data Control Language?

Pick **ONE** option

☐ Revoke

☐ Grant

☒ Both A & B

☐ None of the above

12. Consider the following transaction involving two bank accounts x and y. read (x); x := x - 50 ; write (x) ; read (y) ; y := y + 50 ; write (y) The constraint that the sum of the accounts x and y should remain constant is that of:

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The constraint that the sum of the accounts x and y should remain constant is that of:

Pick **ONE** option

☐ Durability

☒ Consistency

☐ Isolation

☐ Both b) and c)

Clear Selection

9

13. What is the full form of DBaaS?

0

What is the full form of DBaaS?

Pick **ONE** option

☐ Data as a Service

☒ Database as a Service ✓

☐ Database as a System

☐ Data as a System

Clear Selection

15. Which of the following are not true when using a Cache tier in Database design ?

Which of the following are not true when using a Cache tier in Database design ?

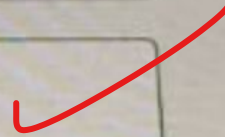
Pick **ONE** option

☐ A cache tier improves system performance.

☐ A cache tier reduces database workloads.

☐ A cache tier can be scaled up independently.

☒ A cache tier increases database workloads.



ar Selection

20. What is the purpose of NAT (Network Address Translation) in networking?

What is the purpose of NAT (Network Address Translation) in networking?

Pick **ONE** option

- ☐ To translate domain names to IP addresses
- ☐ To encrypt network traffic for secure communication
- ☐ To assign IP addresses dynamically to devices on a network
- ☒ To map private IP addresses to public IP addresses for internet access

Selection

23. Date Math

If today is Monday, what day is it after 62 days?

Pick **ONE** option

☒ Sunday

☐ Saturday

☐ Monday

☐ Wednesday

Clear Selection

26. Integer Division

What is the remainder of 1992 divided by 92?

Pick **ONE** option

☐ 0

☐ 1

☐ 40

☐ None of the above



60

27. Logic Puzzles 7

A and B together can do a piece of work in 10 days.
B and C together can do the same piece of work in 5 days.
C and D together can do it in 4 days.
A can do it in 40 days, alone.

Which is the correct statement.

Pick **ONE** option

☐ A and B do the work at the same rate.

☐ B and D do the work at the same rate.

☒ C and D do the work at the same rate.

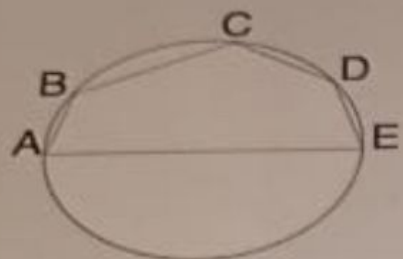
☐ A and D do the work at the same rate.

✓ GPT

29. Sum of Angles

A, B, C, D, E are five points marked in order on the circumference of a semi-circle while traversing clockwise. A and E are the ends of the diameter.

What is the sum of angles **ABC** and **CDE**?



Pick **ONE** option

☐ 135 Degrees

☐ 180 Degrees

☐ 225 Degrees

☐ 315 Degrees

☐ 00 Degrees

30. Set Cardinality 2

Consider a set S that consists of real positive values of x that satisfy the equation $(x^2 + 5x + 4)/(x^2 - 7x + 12) \leq 0$, where $x \neq 3, 4$. What is the cardinality of S ?

Pick **ONE** option

☐ 0

☐ 1

☐ 2

☐ 3

Clear Selection

31. Distance From Start

A hiker walks 50m north, turns left, and walks 30m. The hiker turns left and walks for 50m, then turns left and walks 50m more. How far is the hiker from the starting point?

Pick **ONE** option

☐ 25m

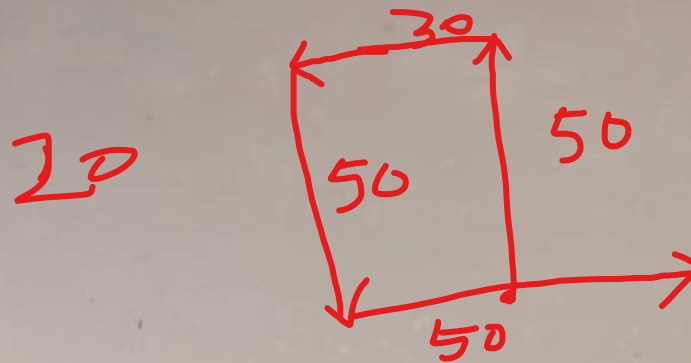
☐ 50m

☐ 35m

☒ None of the above



Clear Selection



32. Digit Sum Reduction

The digit sum of a number is found by repeatedly summing the digits until a single-digit number is reached. Find the digit sum of 25! (25 factorial).

e.g. digit sum of 378 =

$$3 + 7 + 8 = 18$$

$$1 + 8 = 9$$

Pick **ONE** option

☐ 6

☐ 7

☐ 8

☒ 9

ear Selection

1551121004 3330985984 000000

33. Mean Probability Distribution

In a group of 50 people, 20 are vegetarian. If two people are selected at random, what is the expected number of vegetarians?

Pick **ONE** option

☐ 197/245

☐ 198/245

☐ 196/245

☐ 191/245

Clear Selection

34. Ages and Accidents

A man spent $\frac{1}{3}$ rd of his life as a bachelor. After ten years of his marriage a son was born to him. The father and son went on a vacation together at the time the father's age was double that of his son. What was the father's age at the time of the vacation?

Pick **ONE** option

☐ 50

☐ 60

☐ 70

☐ 80

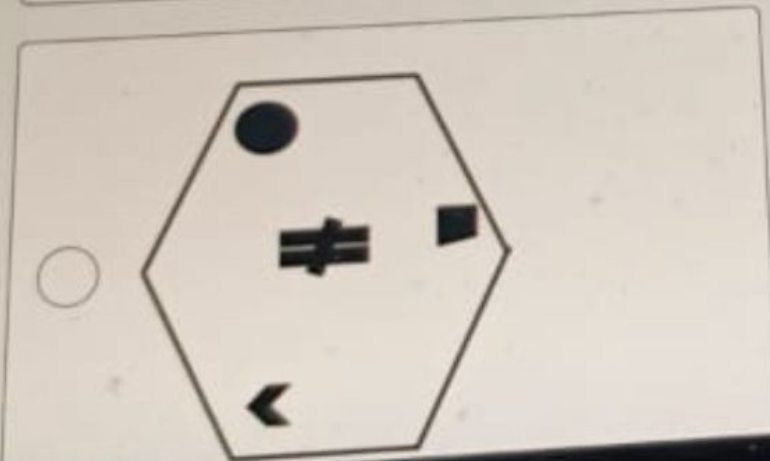
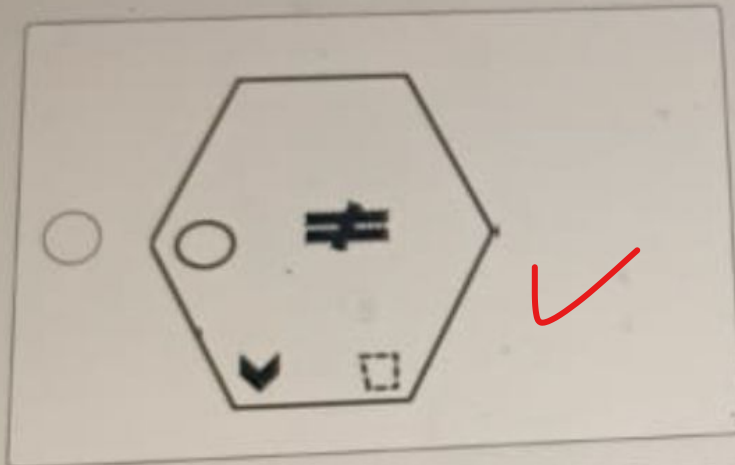
ar Selection

22. Diagrammatic Reasoning 1

Which option will replace the question mark (?)



Pick **ONE** option



✓

✓

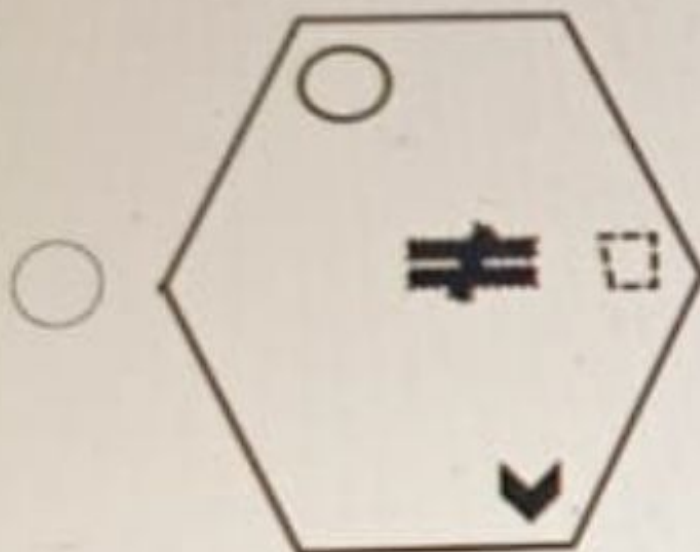
22



✓

✓

25



✓

Clear Selection